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# CPCS-214 Computer Architecture and Organization

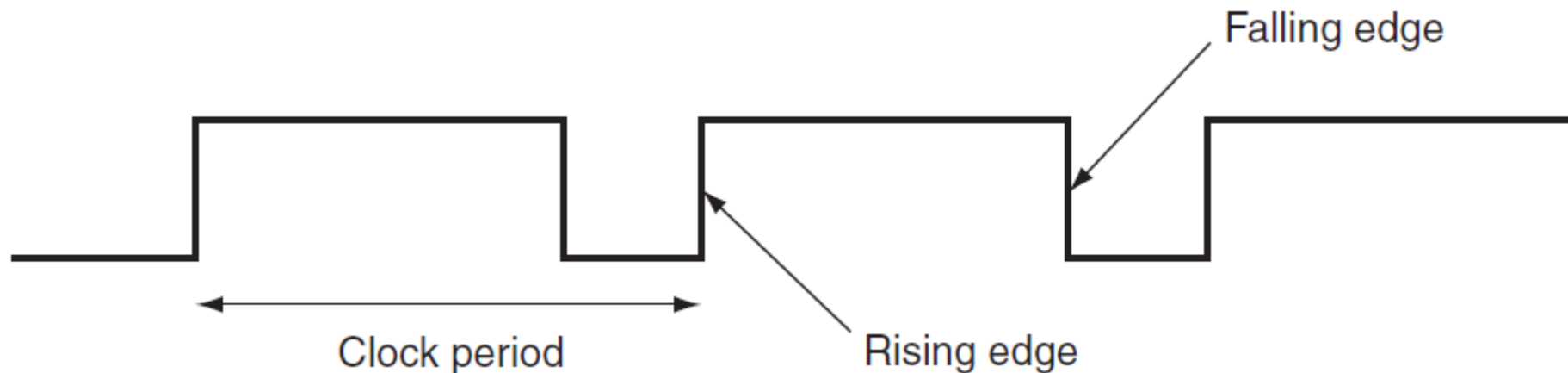
## 33 - Memory Elements

# Introduction

- Registers and register files provide basic building blocks for small memories
- Larger amounts of memory are built using either
  - *SRAMs* (static random access memories) or
  - *DRAMs* (dynamic random access memories)

# Clock

- *Clock cycle time* or *clock period* is divided into two portions: when the clock is *high* and when the clock is *low*



# Edge-triggered Clocking

- We use only **edge-triggered clocking**
  - means that all state changes occur on a clock edge
  - either **rising edge** or the **falling edge** of clock is *active* and causes state changes to occur
- **State elements** in an edge-triggered design are implemented so that contents of state elements only change on active clock edge
- Choice of which edge is active is influenced by implementation technology

# SRAMs

- ICs that are memory arrays with (usually) a single access port that can provide either a read or a write
- Have a fixed access time to any datum, though read and write access characteristics often differ

# SRAM

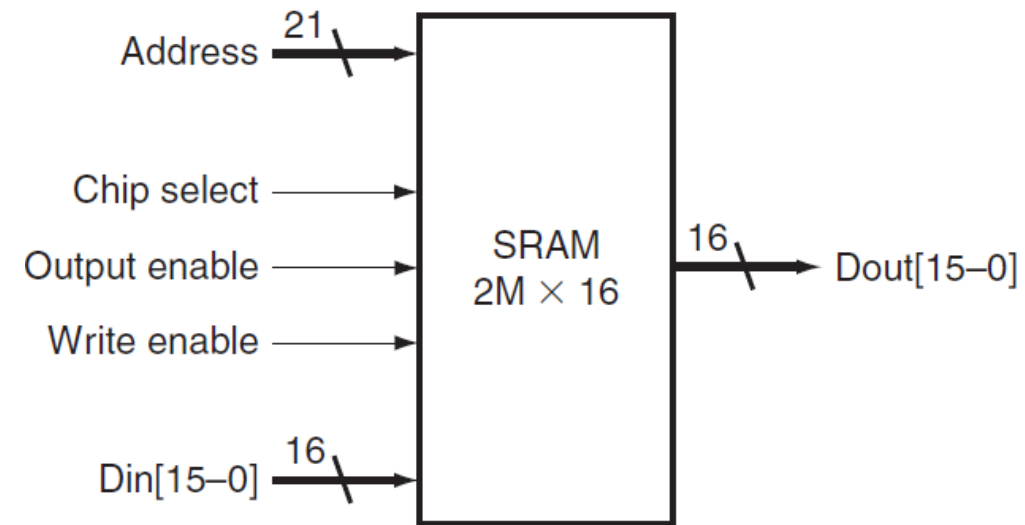
- An SRAM chip has specific configuration in terms of number of addressable locations, as well as width of each addressable location
  - For example, a  $4\text{M} \times 8$  SRAM provides 4M entries, each of which is 8 bits wide
  - Thus it will have 22 address lines (since  $4\text{M} = 2^{22}$ ), an 8-bit data output line, and an 8-bit single data input line

# SRAM

- Number of addressable locations is often called the *height*, with number of bits per unit called *width*
- For variety of technical reasons, newest and fastest SRAMs are typically available in narrow configurations:  $\times 1$  and  $\times 4$
- Figure shows input and output signals for a  $2\text{M} \times 16$  SRAM

# Reads

- To initiate a **read** or **write** access, *Chip select* signal must be made active
  - For reads, must also activate *Output enable* signal that controls datum selected by address driven to pins
  - *Output enable* is useful for connecting multiple memories to a single-output bus and using *Output enable* to determine which memory drives bus



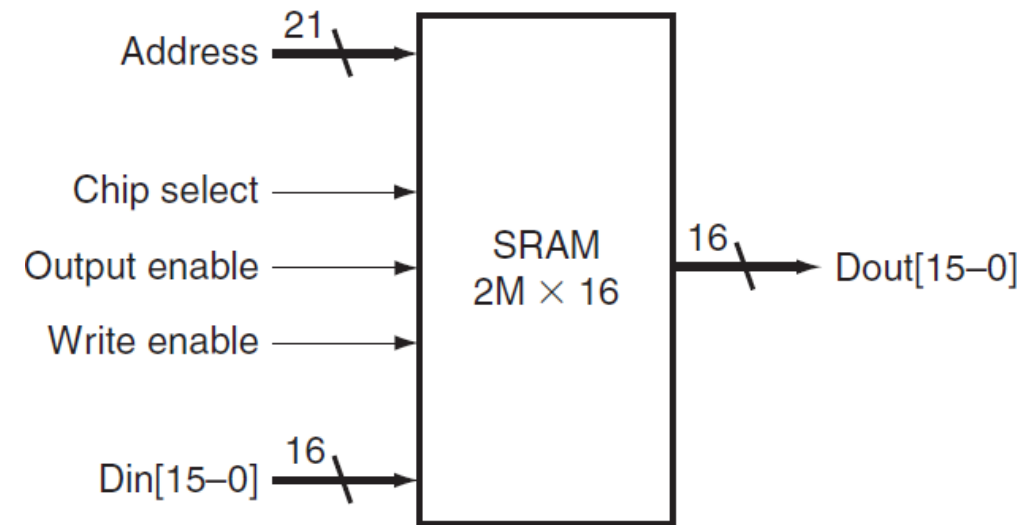


# Access Time

- Delay from time that *Output enable* is true and address lines are valid until time that data is on output line
- Typical read access times varied from about 2–4 ns for the fastest, which tend to be somewhat smaller and narrower, to 8–20 ns for typical largest parts
- Demand for low-power SRAMs for consumer products and digital appliances has grown greatly in the past five years; these SRAMs have much lower stand-by and access power, but usually are 5–10 times slower

# Writes

- Must supply **data** to be written and **address**, as well as signals to cause **write** to occur
- When both *Write enable* and *Chip select* are **true**, data on data input lines is written into the cell specified by address



# DRAMs

- In SRAM, value stored in a cell is kept on a pair of inverting gates, and as long as power is applied, value can be kept indefinitely
- In DRAM, value kept in a cell is stored as a charge in a capacitor; a single transistor is then used to access this stored charge, either to read value or to overwrite charge stored there

# DRAMs

- Because DRAMs use only a single transistor per bit of storage, they are much denser and cheaper per bit
- By comparison, SRAMs require four to six transistors per bit
- Because DRAMs store charge on a capacitor, it cannot be kept indefinitely and must periodically be *refreshed*
  - That is why this memory structure is called *dynamic*, as opposed to static storage in a SRAM cell

# Refreshing

- To refresh cell, read its contents and write it back
- Charge can be kept for several milliseconds, which might correspond to close to a million clock cycles
- Today, single-chip memory controllers often handle the refresh function independently of the processor

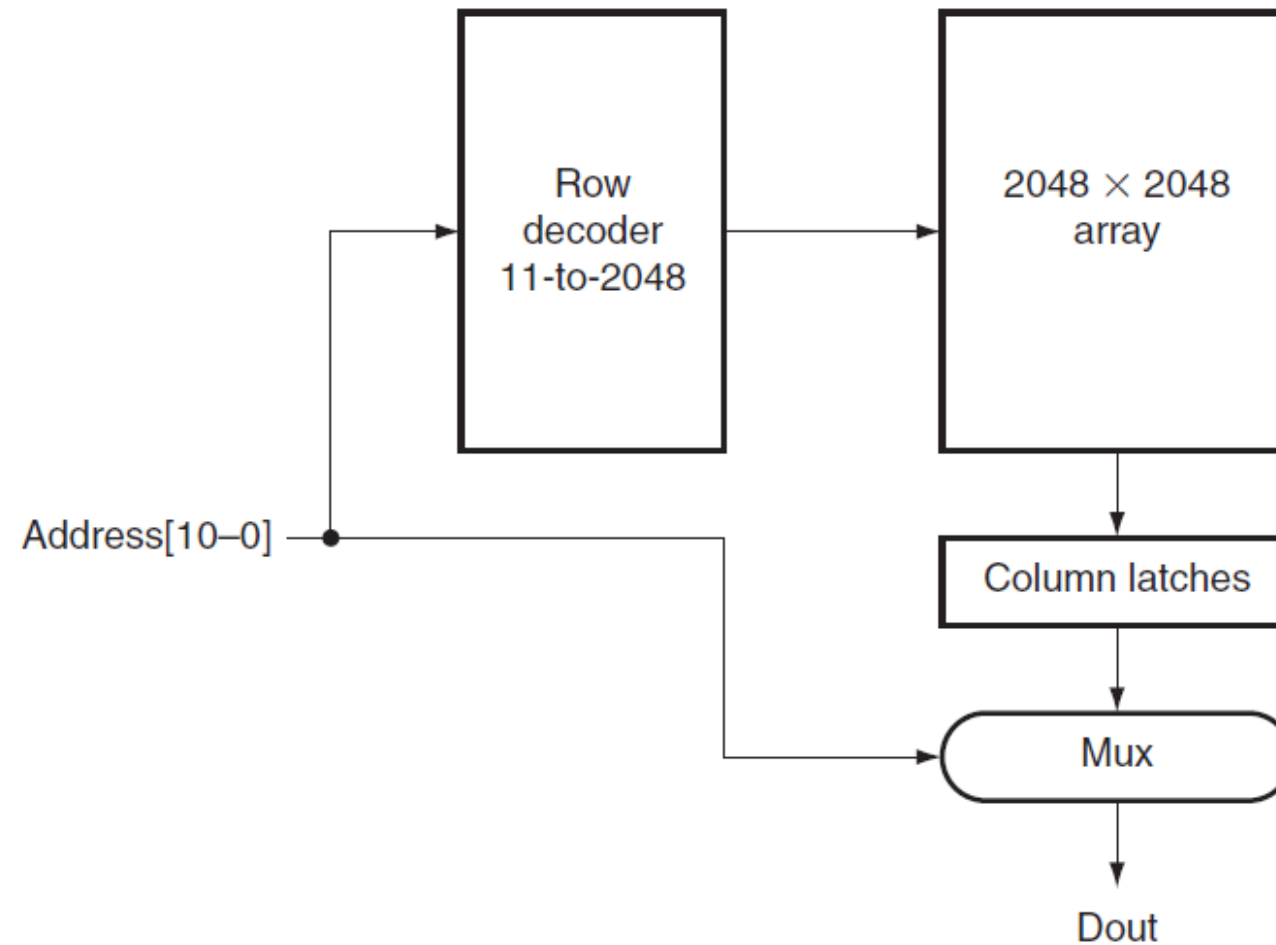
# Refreshing

- If every bit had to be read out then written back individually, with large DRAMs, we would constantly be refreshing DRAM, leaving no time for accessing it
- Fortunately, DRAMs use two-level decoding structure
  - This allows to refresh an entire row (which shares a word line) with a read cycle followed immediately by a write cycle
- Typically, refresh operations consume 1% to 2% of active cycles of DRAM, leaving remaining 98% to 99% of cycles available for reading and writing data

# Two-Level Decoding

- DRAMs use a two-level decoder consisting of a *row access* followed by a *column access*, as shown in figure
  - Row access chooses one of a number of rows and activates the corresponding word line
  - Contents of all columns in active row are then stored in a set of latches
  - Column access then selects data from column latches
- To save pins and reduce package cost, same address lines are used for both row and column address

# Two-Level Decoding



A 4M × 1 DRAM is built with a 2048 × 2048 array



# Refreshing

- Refresh is performed by simply reading columns into column latches and then writing the same values back
- Thus, an entire row is refreshed in one cycle

# Access Time

- The two-level addressing scheme, combined with the internal circuitry, makes DRAM access times much longer (by a factor of 5–10) than SRAM access times
- Typical DRAM access times ranged from 45 to 65 ns
- The much lower cost per bit makes DRAM the choice for main memory, while the faster access time makes SRAM the choice for caches

# SDRAM

- DRAM designers have used the internal structure of DRAM as a way to provide higher bandwidth out of a DRAM
  - Done by allowing column address to change without changing row address, resulting in an access to other bits in column latches
- To make this process faster and more precise, address inputs were clocked, leading to the dominant form of DRAM in use today: **synchronous DRAM** or **SDRAM**
  - SDRAMs have been memory chip of choice for most cache-based main memory systems

# DDRRAMs

- DDRRAMs (Double Data Rate RAMs)
- Called **double data rate** because they transfer data on both the **rising** and **falling edge** of an externally supplied clock, were the most heavily used form of SDRAMs
- These high-speed transfers can be used to boost bandwidth available out of main memory to match needs of processor and caches